Cloud Computing

Lab-3

Name: Monisha Sharma
SRN:PES2UG23CS906
Class: 6F
Date:29/1/2026

SS1

```
mona@MonasLaptopy:~/PES2UG23CS906$ export SRN=PES2UG23CS906
mona@MonasLaptopy:~/PES2UG23CS906$ echo "SRN: $SRN"
SRN: PES2UG23CS906
mona@MonasLaptopy:~/PES2UG23CS906$ docker run hello-world

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/

mona@MonasLaptopy:~/PES2UG23CS906$
```

SS2

```
mona@MonasLaptopy:~/PES2UG23CS906$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
mona@MonasLaptopy:~/PES2UG23CS906$ docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                      PORTS     NAMES
9a41111e425e   hello-world   "/hello"   54 seconds ago   Exited (0) 53 seconds ago             jolly_bhabha
3f0938401ab0   hello-world   "/hello"   3 minutes ago    Exited (0) 3 minutes ago              unruffled_shannon
mona@MonasLaptopy:~/PES2UG23CS906$ docker images
                                                                                                  i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   05813aedc15f       25.9kB         9.52kB    U
mona@MonasLaptopy:~/PES2UG23CS906$
```

SS3

```
mona@MonasLaptopy:~/PES2UG23CS906$ docker pull python:3.10-slim
3.10-slim: Pulling from library/python
ed8cfd59ed51: Pull complete
47c2bec0e44b: Pull complete
af233a02f79e: Pull complete
119d43eec815: Pull complete
f81c64c75055: Download complete
b3e684e4d1bb: Download complete
Digest: sha256:f5d029fe39146b08200bcc73595795ac19b85997ad0e5001a02c7c32e8769efa
Status: Downloaded newer image for python:3.10-slim
docker.io/library/python:3.10-slim
mona@MonasLaptopy:~/PES2UG23CS906$ docker images

i Info →   U  In Use
IMAGE                  ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest     05813aedc15f       25.9kB         9.52kB    U
python:3.10-slim       f5d029fe3914        185MB         47.1MB
mona@MonasLaptopy:~/PES2UG23CS906$
```

SS4

EXPLORER

ML-DOCKER-PES2UG23CS906
- templates
  - index.html
- app.py
- Dockerfile
- spam_classifier_model.pkl
- tfidf_vectorizer.pkl

Dockerfile | index.html

Dockerfile

```
# Step 1: Specify the base image (use Python slim version)
FROM python:3.11-slim

# Step 2: Set the working directory inside the container
WORKDIR /app

# Step 3: Copy the application file into the container
COPY app.py ./

# Step 4: Copy the ML model file into the container
COPY spam_classifier_model.pkl ./

# Step 5: Copy the vectorizer file into the container
COPY tfidf_vectorizer.pkl ./

# Step 6: Copy the HTML templates folder
COPY templates ./templates

# Step 7: Install required Python libraries ( you require flask , scikit-learn and joblib)
RUN pip install --no-cache-dir flask scikit-learn joblib

# Step 8: Expose the port used by the Flask app
EXPOSE 5000

# Step 9: Command to start the application when container runs
CMD ["python", "app.py"]
```

```
mona@MonasLaptopy:~/PES2UG23CS906$ mkdir ml-docker-PES2UG23CS906
mona@MonasLaptopy:~/PES2UG23CS906$ cd ml-docker-PES2UG23CS906
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ ls
app.py  spam_classifier_model.pkl  templates  tfidf_vectorizer.pkl
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker build -t spam-web-app-pes2ug23cs906 .
[+] Building 54.8s (12/12) FINISHED                                                                                docker:default
 => [internal] load build definition from Dockerfile                                                                          0.0s
 => => transferring dockerfile: 829B                                                                                          0.0s
 => [internal] load metadata for docker.io/library/python:3.11-slim                                                           1.6s
 => [internal] load .dockerignore                                                                                             0.0s
 => => transferring context: 2B                                                                                               0.0s
 => [1/7] FROM docker.io/library/python:3.11-slim@sha256:5be45dbade29bebd6886af6b438fd7e0b4eb7b611f39ba62b430263f82de36d2       0.1s
 => => resolve docker.io/library/python:3.11-slim@sha256:5be45dbade29bebd6886af6b438fd7e0b4eb7b611f39ba62b430263f82de36d2       0.1s
 => [internal] load build context                                                                                             0.0s
 => => transferring context: 184B                                                                                             0.0s
 => CACHED [2/7] WORKDIR /app                                                                                                 0.0s
 => CACHED [3/7] COPY app.py ./                                                                                               0.0s
 => [4/7] COPY spam_classifier_model.pkl ./                                                                                   0.1s
 => [5/7] COPY tfidf_vectorizer.pkl ./                                                                                        0.1s
 => [6/7] COPY templates ./templates                                                                                          0.1s
 => [7/7] RUN pip install --no-cache-dir flask scikit-learn joblib                                                           27.4s
 => exporting to image                                                                                                       24.9s
 => => exporting layers                                                                                                      20.2s
 => => exporting manifest sha256:aa18ce5f79ec4e5f7204f68c8549e4d32001e4f2244b180c9e68175344ba4dee                              0.0s
 => => exporting config sha256:a969c1cbaed866a9c9a4e4c05e8e4197698249c8ff6357943248f24c005d31e7                               0.0s
 => => exporting attestation manifest sha256:a187955c512567feab4104f5f8fd3d29f9c2699b96b150fde6afe07d43a13a58                 0.1s
 => => exporting manifest list sha256:c5e6ac77a45f7d3320b143b90c6eef06e0ba29601ceb277b36e7f4816cdfec70                         0.1s
 => => naming to docker.io/library/spam-web-app-pes2ug23cs906:latest                                                          0.0s
 => => unpacking to docker.io/library/spam-web-app-pes2ug23cs906:latest                                                       4.3s
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

Latest version of docker doesn’t consider CMD and EXPORT as build and shows the built successfuloly by saying “FINISHED”

SS5

```
 => => naming to docker.io/library/spam-web-app-pes2ug23cs906:latest                                                          0.0s
 => => unpacking to docker.io/library/spam-web-app-pes2ug23cs906:latest                                                       4.3s
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker run -d -p 5000:5000 spam-web-app-pes2ug23cs906
999053e49d075e30b89d2550f8a7a3c9debe4ab30ecd17312e985ffd6b5076ce
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker ps
CONTAINER ID   IMAGE                       COMMAND           CREATED          STATUS          PORTS
        NAMES
999053e49d07   spam-web-app-pes2ug23cs906   "python app.py"   20 seconds ago   Up 19 seconds   0.0.0.0:5000->5000/tcp, [::]:5000->500
0/tcp   objective_pascal
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS6

EXPLORER

ML-DOCKER-PES2UG23CS906

- templates
  - index.html
- app.py
- Dockerfile
- spam_classifier_model.pkl
- tfidf_vectorizer.pkl

Dockerfile | index.html

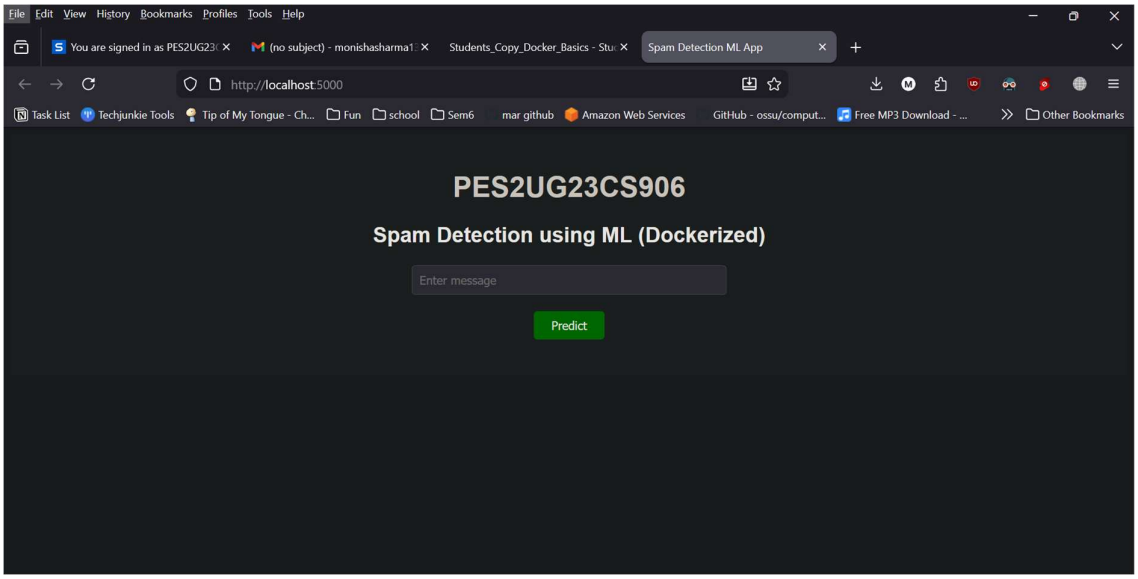
templates > index.html > ...

```html
<!DOCTYPE html>
<html>
<head>
    <title>Spam Detection ML App</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            padding: 20px;
            background-color: #f9f9f9;
        }
        h1, h2, h3 {
            text-align: center;
        }
        h1 {
            color: #333;
        }
        form {
            text-align: center;
            margin: 20px 0;
        }
        input[type="text"] {
            padding: 8px;
            border: 1px solid #ccc;
            border-radius: 4px;
        }
        button {
            padding: 8px 20px;
            background-color: green;
            color: white;
            border: none;
            cursor: pointer;
            border-radius: 4px;
        }
        button:hover {
            background-color: darkgreen;
        }
        h3 {
            background-color: #e8f5e9;
            padding: 10px;
            border-radius: 4px;
        }
    </style>
</head>
<body>
    <div class="container">
        <h1> PES2UG23CS906</h1>
        <h2>Spam Detection using ML (Dockerized)</h2>

    <form method="POST">
        <input type="text" name="message" placeholder="Enter message" size="45" required>
```

OUTLINE

TIMELINE



SS7

```
0/tcp   objective_pascal
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker login
Authenticating with existing credentials... [Username: pes2ug23cs906]

i Info → To login with a different account, run 'docker logout' followed by 'docker login'


Login Succeeded
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

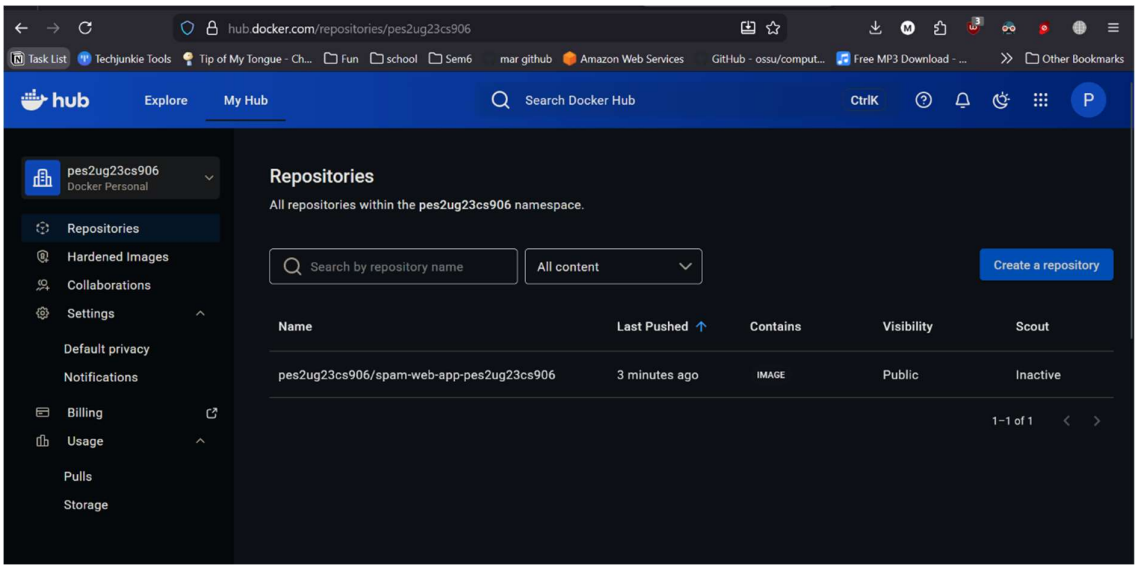
SS8

```
Login Succeeded
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker tag spam-web-app-pes2ug23cs906 pes2ug23cs906/spam-web-app-pes2ug23c
s906:v1
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                                  i Info →   U  In Use
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                             05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1    c5e6ac77a45f        570MB          132MB    U
python:3.10-slim                               f5d029fe3914        185MB         47.1MB
spam-web-app-pes2ug23cs906:latest              c5e6ac77a45f        570MB          132MB    U
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS9

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker push pes2ug23cs906/spam-web-app-pes2ug23cs906:v1
The push refers to repository [docker.io/pes2ug23cs906/spam-web-app-pes2ug23cs906]
fb375704384a: Pushed
119d43eec815: Mounted from library/python
49a8dffbacc4: Pushed
e7af7557d64a: Pushed
3e731abb5c1d: Pushed
b9cc728f39eb: Pushed
5b09819094bb: Pushed
13a4bc30c803: Pushed
0b2bf04f68e9: Pushed
5184ab728e28: Pushed
f59a325c0f48: Pushed
v1: digest: sha256:c5e6ac77a45f7d3320b143b90c6eef06e0ba29601ceb277b36e7f4816cdfec70 size: 856
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS10



SS11

```
v1: digest: sha256:c5e6ac77a45f7d3320b143b90c6eef06e0ba29601ceb277b36e7f4816cdfec70 size: 856
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker ps
CONTAINER ID   IMAGE                          COMMAND           CREATED         STATUS         PORTS
      NAMES
999053e49d07   spam-web-app-pes2ug23cs906   "python app.py"   7 minutes ago   Up 7 minutes   0.0.0.0:5000->5000/tcp, [::]:5000->5000/
tcp   objective_pascal
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker stop 999053e49d07
999053e49d07
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS12

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker ps -a
CONTAINER ID   IMAGE                          COMMAND           CREATED         STATUS                       PORTS     NAMES
999053e49d07   spam-web-app-pes2ug23cs906   "python app.py"   8 minutes ago   Exited (137) 50 seconds ago             objective_pasc
al
9a41111e425e   hello-world                    "/hello"          45 minutes ago  Exited (0) 45 minutes ago               jolly_bhabha
3f0938401ab0   hello-world                    "/hello"          48 minutes ago  Exited (0) 48 minutes ago               unruffled_shan
non
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rm 999053e49d07
999053e49d07
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker ps -a
CONTAINER ID   IMAGE         COMMAND    CREATED          STATUS                     PORTS     NAMES
9a41111e425e   hello-world   "/hello"   46 minutes ago   Exited (0) 46 minutes ago            jolly_bhabha
3f0938401ab0   hello-world   "/hello"   49 minutes ago   Exited (0) 49 minutes ago            unruffled_shannon
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS13

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                                       ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                          05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1 c5e6ac77a45f        570MB          132MB
python:3.10-slim                            f5d029fe3914        185MB         47.1MB
spam-web-app-pes2ug23cs906:latest           c5e6ac77a45f        570MB          132MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rmi c5e6ac77a45f
Error response from daemon: conflict: unable to delete c5e6ac77a45f (must be forced) - image is referenced in multiple repositories
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rmi spam-web-app-pes2ug23cs906:latest
Untagged: spam-web-app-pes2ug23cs906:latest
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                                       ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                          05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1 c5e6ac77a45f        570MB          132MB
python:3.10-slim                            f5d029fe3914        185MB         47.1MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

SS14

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                             05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1    c5e6ac77a45f        570MB          132MB
python:3.10-slim                               f5d029fe3914        185MB         47.1MB
spam-web-app-pes2ug23cs906:latest              c5e6ac77a45f        570MB          132MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rmi c5e6ac77a45f
Error response from daemon: conflict: unable to delete c5e6ac77a45f (must be forced) - image is referenced in multiple repositories
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rmi spam-web-app-pes2ug23cs906:latest
Untagged: spam-web-app-pes2ug23cs906:latest
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                             05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1    c5e6ac77a45f        570MB          132MB
python:3.10-slim                               f5d029fe3914        185MB         47.1MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker rmi pes2ug23cs906/spam-web-app-pes2ug23cs906:v1
Untagged: pes2ug23cs906/spam-web-app-pes2ug23cs906:v1
Deleted: sha256:c5e6ac77a45f7d3320b143b90c6eef06e0ba29601ceb277b36e7f4816cdfec70
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest   05813aedc15f       25.9kB         9.52kB    U
python:3.10-slim     f5d029fe3914        185MB         47.1MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```

```
l.t   msg3.txt   msg2.txt   lab0_cc.txt   https p   research.txt   Genai_prog_   pes2u
File   Edit   View                                  H1

pes2ug23cs906

Yes, Docker provides commands to delete images completely in one go.

The command docker image prune -a removes all unused Docker images, including images with multiple tags, as long as
they are not used by any container.

Another command docker rmi -f $(docker images -q) forcefully deletes all Docker images at once by removing every
image ID from the system.

Normally, deleting a single tag does not remove the image if other tags still reference the same image ID. These
commands ensure complete removal of images.
```

SS15

```
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker pull pes2ug23cs906/spam-web-app-pes2ug23cs906:v1

v1: Pulling from pes2ug23cs906/spam-web-app-pes2ug23cs906
13a4bc30c803: Pull complete
e7af7557d64a: Pull complete
49a8dffbacc4: Pull complete
b9cc728f39eb: Pull complete
5184ab728e28: Pull complete
f59a325c0f48: Pull complete
fb375704384a: Download complete
Digest: sha256:c5e6ac77a45f7d3320b143b90c6eef06e0ba29601ceb277b36e7f4816cdfec70
Status: Downloaded newer image for pes2ug23cs906/spam-web-app-pes2ug23cs906:v1
docker.io/pes2ug23cs906/spam-web-app-pes2ug23cs906:v1
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$ docker images
                                                                                                    i Info →
IMAGE                                          ID             DISK USAGE   CONTENT SIZE   EXTRA
hello-world:latest                             05813aedc15f       25.9kB         9.52kB    U
pes2ug23cs906/spam-web-app-pes2ug23cs906:v1    c5e6ac77a45f        570MB          132MB
python:3.10-slim                               f5d029fe3914        185MB         47.1MB
mona@MonasLaptopy:~/PES2UG23CS906/ml-docker-PES2UG23CS906$
```